

Sachin Bala

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Technical Skills

Python (Pandas, NumPy, scikit-learn, statsmodels), SQL (MySQL), R, Statistical Modeling, Machine Learning, A/B Experimentation, DOE, SPC, Process Capability Studies, Anomaly Detection, Data Visualization (Grafana, Matplotlib), Computer Vision (YOLO), JMP, Minitab, LabVIEW, SolidWorks, Six Sigma, PFMEA, Lean Manufacturing, Microsoft Suite, Cognex, Dr Schenk vision system

Education

Western Michigan University (WMU)

Overall GPA: 3.69

Bachelor's in Mechanical Engineering with minor in Computer Science and Mathematics

Work Experiences

Associate Process Development Engineer, Tesla Inc., Austin, TX

02/25-present

- Translated complex business and operational problems into precise data requirements; built end-to-end statistical models and ML pipelines in Python, R, MySQL, and JMP to optimize product performance and drive measurable improvements in yield and throughput.
- Applied rigorous experimental design (DOE, SPC, capability studies, 8D, 5-Why, FMEA) to large-scale manufacturing datasets; identified root causes of performance excursions and delivered quantitative recommendations that directly improved operational efficiency.
- Commissioned two electrode production lines achieving 98% yield through systematic data-driven experimentation and qualification; defined and monitored real-time operational metrics to guide investment and product decisions at the factory level.

Engineering Technician, Process Development Team, Tesla Inc., Austin, TX

8/23-02/25

- Designed and deployed a scalable real-time data pipeline in Grafana tracking tool performance metrics across the production floor; automated anomaly detection reduced unplanned failures and enabled proactive corrective action.
- Used Python scripting and statistical analysis to identify process improvements from large operational datasets, achieving a 10% increase in production rate while sustaining quality targets.
- Conducted structured root cause analysis on downtime events using SPC charts, fault logs, and process historians; quantified impact and implemented corrective actions that improved operational uptime.

Engineering Co-Op, Scherdel SST, Muskegon, MI

8/22-12/22

- Analyzed EV component market data and manufacturing material trends to support product launch decisions and quality process design; delivered quantitative findings to cross-functional stakeholders.

Process Engineering Intern, Scherdel SST, Muskegon, MI

05/22-08/22

- Built a predictive failure-detection model using ML algorithms and Python-based statistical analysis, improving manufacturing efficiency and accuracy by 25%; validated with Gauge R&R methodology and F1-score metrics.
- Developed a Computer Vision defect detection system; rigorously validated model performance using structured quantitative methods (Gauge R&R, F1-score analysis).

Laser Tech Technician Internship, Technique Inc., Jackson, MI

09/21-04/22

- Optimized laser cutting using CAD and machine troubleshooting; validated chassis parts via design review and laser tracker measurement data.
- Updated SOPs and trained technicians and operators to follow standardized procedures.

Engineering Research and Project

Senior Design Project - Two Tier Industrial Cart (Technique Inc.)

01/22-12/22

- Designed a two-tier industrial cart in SolidWorks with scissor lift mechanism; validated with FEA analysis and numerical calculations.
- Applied DFMEA to identify and mitigate risk, establishing data-backed design controls and a quality control plan.

Honors and Membership

Diether H. Haenicke Scholarship, WMU Dean's List, Yellow Belt Training (Six Sigma), Root Nepal (Co-Founder)